Last name _____

First name _____

LARSON—MATH 310–HOMEWORK WORKSHEET 12
Least Squares Data Fitting

1. Write up a **neat** assignment on a **new sheet** of paper. (Do not cram your answers between the lines).
2. **Number** your problems so that it is easy to see what work matches the assigned problems.
3. Be verbose. Remember that you do not understand a concept if you do not know an **examples**.

Problems

1. **Data:** Points $(-1, 1), (0, 0), (1, 2)$ in $\mathbb{R}^2$.

   Goal: Find the least-squares best fit line for this data. Assume our model is *linear* of the form $y = b + mx$, where b is the y-intercept of the line and m is the slope. Ideally we'd want each point (x_i, y_i) to satisfy:

   $$y_i = b + mx_i,$$

   (a) Write out the desired equations.

   (b) Argue that no linear function (no values of b and m) can exactly satisfy this data.

   (c) Write these equations in matrix form $A \begin{bmatrix} b \\ m \end{bmatrix} = \vec{y}$.

   (d) What is A?

   (e) Check that A has linearly independent columns.

   (f) Find the pseudo-inverse $A^\dagger$ of A.

   (g) Let $\tilde{x}$ be the least squares best fit solution (given in terms of $A^\dagger$). Find $\tilde{x}$.

   (h) What is the corresponding linear function?

   (i) Use it to find an estimated y when $x = 3$.

(over)

2. **Data:** Points $(2,1),(3,0),(5,7)$ in $\mathbb{R}^2$.

   Goal: Find the least-squares best fit line for this data. Assume our model is *linear* of the form $y = b + mx$, where b is the y-intercept of the line and m is the slope. Ideally we'd want each point (x_i, y_i) to satisfy:

$$y_i = b + mx_i,$$

   (a) Write out the desired equations.

   (b) Argue that no linear function (no values of b and m) can exactly satisfy this data.

   (c) Write these equations in matrix form $A \begin{bmatrix} b \\ m \end{bmatrix} = \vec{y}$.

   (d) What is A?

   (e) Check that A has linearly independent columns.

   (f) Find the pseudo-inverse $A^\dagger$ of A.

   (g) Let $\tilde{x}$ be the least squares best fit solution (given in terms of $A^\dagger$). Find $\tilde{x}$.

   (h) What is the corresponding linear function?

   (i) Use it to find an estimated y when $x = 10$.